

Day 5 Homework: A second document type

Phase 1 · Week 1 · Day 5 · Sadasivam Mithran · Repository: github.com/mithransadasivam/FDE (folder HW #5)

Schema

IncidentReport in `app/schemas.py`, next to the unchanged **Invoice**. Severity is a `Literal` of P1 to P4; the incident ID must match `INC-YYYY-NNNN`; `resolved_at` and `root_cause` are optional because an open incident has neither (report 02); `actions` needs at least one item. The time-order rule is checked in code. One addition beyond the brief: if only one of the two times has a timezone, validation fails, because comparing them would otherwise raise a `TypeError` instead of a validation error, and a timezone is never guessed.

```
class IncidentReport(BaseModel):
    incident_id: str = Field(pattern=r"^INC-\d{4}-\d{4}$")
    system: str = Field(min_length=2)
    severity: Literal["P1", "P2", "P3", "P4"]
    started_at: datetime
    resolved_at: datetime | None = None # open incidents have no resolution time
    impact: str
    root_cause: str | None = None # still under investigation on open incidents
    actions: list[str] = Field(min_length=1)

    @model_validator(mode="after")
    def resolved_after_started(self):
        if self.resolved_at is None:
            return self
        # an aware and a naive datetime cannot be compared; reject rather than guess a zone
        if (self.resolved_at.tzinfo is None) != (self.started_at.tzinfo is None):
            raise ValueError(
                "started_at and resolved_at must both have a timezone or both have none"
            )
        if self.resolved_at < self.started_at:
            raise ValueError("resolved_at is earlier than started_at")
        return self
```

```
SCHEMAS = {"invoice": Invoice, "incident": IncidentReport}
```

Results

Command: `python -m scripts.extract_batch --retry --type incident --folder data/incidents` (new `--type` option, default `invoice`; incidents show severity and system instead of the total).

document	result	tries	detail
Day05_HW_Slide03_incident_01.txt	VALID	1	P1 Payments API
Day05_HW_Slide03_incident_02.txt	VALID	1	P3 ORD-ETL
Day05_HW_Slide03_incident_03.txt	VALID	1	P2 Wi-Fi

3 of 3 documents valid

Each result compared with its source report:

Report	Result	Check against the source
01 · Payments API	Valid · P1	Both times match (14:05 and 15:20, +05:30), resolved after started. Impact and root cause match. The two actions are the two in the report.
02 · ORD-ETL	Valid · P3	Still open: resolved_at and root_cause are null , nothing guessed. Started 01:40 on 21 Sep. Both actions match; none added.
03 · Wi-Fi, 3rd floor	Valid · P2	25/09/2026 read as 25 September (09:15 to 11:40). ID <code>INC-2026-0925</code> found on the first line. Both actions match; none added.

Errors found by comparing with the source: no wrong values. Two small losses: report 03's system is "Wi-Fi" and drops "3rd floor, Bangalore office" (the schema has no location field); report 02's `impact` says the finance report was delayed by 90 minutes but leaves out that the job failed twice with a connection timeout. All three passed on the first attempt, so the retry was not used. Validation proves the shape and the rules; only the comparison with the source proves the values.

Tests

tests/test_incident.py, no model and no API calls. Each failure test starts from a valid report and breaks one rule: severity "High", resolved_at before started_at (matched on the error message), and incident ID "0917". The brief asks for four tests; there are six, because I added a test for an open incident and one for the mixed-timezone rule.

```
tests/test_incident.py::test_a_valid_incident_passes PASSED [ 16%]
tests/test_incident.py::test_unknown_severity_is_rejected PASSED [ 33%]
tests/test_incident.py::test_resolved_before_started_is_rejected PASSED [ 50%]
tests/test_incident.py::test_badly_formed_incident_id_is_rejected PASSED [ 66%]
tests/test_incident.py::test_open_incident_needs_no_resolved_at_or_root_cause PASSED [ 83%]
tests/test_incident.py::test_mixed_timezone_is_rejected PASSED [100%]
===== 6 passed in 0.18s =====
```

Seeing the test fail: I temporarily removed only the resolved_at < started_at check. The time-order test failed and the other five passed; I then restored the rule and confirmed with git that the file matched the commit.

```
> with pytest.raises(ValidationError, match="earlier than started_at"):
E     Failed: DID NOT RAISE ValidationError
```

```
tests\test_incident.py:31: Failed
FAILED tests/test_incident.py::test_resolved_before_started_is_rejected
1 failed, 5 passed in 0.30s
```

App and review

app/streamlit_app.py has a Document type selector (Invoice or Incident) built from SCHEMAS; the chosen schema is passed to extract_with_retry and the messages name the chosen type. The type is stored with the result, so changing the selector later cannot relabel an earlier result. A document still invalid after one retry is shown for a person to review, not corrected automatically. The security-reviewer subagent from Day 2 was not available in this repository, so I ran Claude Code's built-in security review on the change instead; it found no vulnerabilities (the schema comes from a fixed list, not user input, and results are rendered with Streamlit's escaping).

Reflection

Next document type: IT support tickets from the helpdesk queue, extracted into fields such as ticket ID, requester's department, category, priority and the stated problem.

The one rule its schema must enforce: priority must be exactly one of the allowed levels (for example P1 to P4), never a free-text word like "urgent". A ticket with a made-up or missing priority would be routed wrongly or missed by the on-call team, so a value outside the list must fail validation and go to a person, as the severity does for incidents.